

A Modulus Proof of Cannon's Conjecture

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Abstract

We prove that every hyperbolic group whose boundary is homeomorphic to the two-sphere admits a proper cocompact isometric action on hyperbolic three-space with finite kernel. This resolves Cannon's conjecture positively.

Contents

1	Introduction	2
2	Boundary geometry and the modulus criterion	4
3	A scalar limit from unbounded modulus	8
4	Exponential records: comparing two magnifications	14
5	A probability flow in the subexponential case	18
6	Same-depth expansions and the measure comparison	24
7	Completion of the proof	29
8	Consequences	29

1 Introduction

Cannon’s conjecture asks whether a hyperbolic group whose boundary is a topological two-sphere acts geometrically on hyperbolic three-space. It links the topology at infinity of a group to its three-dimensional geometry. Here a group is *hyperbolic* if a Cayley graph for a finite generating set has uniformly thin geodesic triangles; its boundary consists of asymptotic classes of geodesic rays, with the usual visual topology. An isometric action is *geometric* if it is proper and cocompact.

The following theorem resolves Cannon’s conjecture positively.

Theorem 1.1 (Cannon’s conjecture). *Let G be a hyperbolic group with ∂G homeomorphic to $\mathbb{S}^2$. There is a homomorphism $\rho : G \rightarrow \text{Isom}(\mathbb{H}^3)$ for which the action on $\mathbb{H}^3$ is proper and cocompact. Its kernel is finite.*

Properness means that $\{g \in G : \rho(g)K \cap K \neq \emptyset\}$ is finite for every compact $K \subset \mathbb{H}^3$; cocompactness means that one compact set has translates covering $\mathbb{H}^3$. A finite kernel is allowed, and the image may contain orientation-reversing isometries. In particular, the theorem does not require a choice of orientation preserved by G .

For torsion-free G , the quotient is a closed hyperbolic three-manifold, possibly nonorientable. We prove this consequence and the resulting virtual-structure statements in [Section 8](#).

History and the classical reduction. Cannon’s 1994 combinatorial Riemann mapping theorem grew out of the problem of finding an analytic structure on a sphere supplied only with combinatorial boundary data. It recovers quasiconformal coordinates from discrete annular moduli [[10](#), pp. 155–160]. In their 1998 paper, Cannon–Swenson stated the group-boundary conjecture and related a geometric action to conformality of the canonical boundary coverings [[14](#), Theorem 5.1 and Conjecture 5.1]. Cannon–Floyd–Parry showed in 1999 that it suffices to find, around each point in every prescribed neighborhood, an annulus whose discrete moduli for separating curves stay bounded away from zero [[12](#), Theorem 8.2]. Their finite subdivision rules provide a related framework for recursive planar tilings [[13](#)]. These works turn the topological problem into quantitative control of crossings at small scales.

Bonk–Kleiner’s uniformization theory gives a metric formulation of this program [[6](#)]. We use the precise criterion in Bourdon–Kleiner [[8](#), Corollary 3.5]: on an approximately self-similar metric sphere, a uniform bound for the combinatorial 2-modulus of all continuous curves of a fixed sufficiently small minimum diameter yields a quasi-Möbius parametrization by the round sphere. The visual boundary has the required self-similarity. Thus the remaining analytic task is an upper modulus bound for its fixed-scale coverings. The criterion and its hypotheses are verified in [Theorem 2.5](#).

A related route uses *Ahlfors regular conformal dimension*: the infimum of the Hausdorff dimensions of Ahlfors regular metrics quasisymmetric to the visual metric. This is a version of the boundary invariant introduced by Pansu [[20](#)]. Bonk–Kleiner proved Cannon’s conjecture when that infimum is attained [[7](#), Theorem 1.1]; their argument connects the minimizing metric to quantitative curve families through the Loewner property [[7](#), Theorem 1.3]. The existence of a minimizing metric is a hypothesis of that criterion.

The algebraic approaches connect surface subgroups, cubulations, and Kleinian realizations. Kahn–Marković constructed almost geodesic surface subgroups in closed orientable hyperbolic three-manifold groups [[18](#), Theorems 1.1–1.2]. Bergeron–Wise used this abundance to cubulate those groups, and more generally proved that quasiconvex codimension-one subgroups separating every distinct boundary pair yield a proper cocompact cubical action [[5](#), Theorem 1.4 and Section

5]. Agol proved virtual specialness for hyperbolic groups with such an action [1, Theorem 1.1]; these results give the consequences recorded in Section 8 once a hyperbolic three-manifold realization is known.

In the recognition direction, Marković showed that enough quasiconvex surface subgroups separating every boundary pair give a Kleinian realization for groups whose sphere-boundary action is faithful and orientation-preserving [19, Theorem 1.1]. Haïssinsky established a virtual Kleinian criterion for cubulated hyperbolic groups with planar boundary [16, Theorem 1.10]. More recently, Groves, Haïssinsky, Manning, Osajda, Sisto, and Walsh constructed group-theoretic drillings and showed that the Toral Relative Cannon conjecture implies a virtual Kleinian realization for every residually finite hyperbolic group with sphere boundary [15, Theorems A–B and Corollary C]. The present proof follows the analytic route, establishing the required modulus bound directly from hyperbolicity and the topology of the boundary.

After boundary uniformization, the invariant-conformal-structure method of Sullivan and the detailed circumcenter construction of Tukia straighten the action [23, 24]. We state the analytic inputs at their point of use and include the short argument with unoriented conformal structures, so that orientation-reversing elements are retained. Properness and cocompactness then follow from the boundary action on ordered triples (Proposition 2.6).

The proof in outline. Fix a visual metric on $Z = \partial G$. At scale a^{-n} , where $a > 1$, cover Z by balls centered on a separated net. Assign nonnegative weights to the balls so that every path of a fixed minimum diameter meets balls of total weight at least one. Let $M(n)$ be the least sum of squared weights. The classical criterion reduces the theorem to

$$\sup_n M(n) < \infty.$$

Assume this fails. Record indices permit comparison of $M(N - l)$ with $M(N)$, and lead to two growth cases.

First, crossing duality in a coordinate rectangle produces weights with squared sum $O(M(N)^{-1})$. A planar regularization turns them into continuous functions H_N on a slightly enlarged rectangle, with separating level continua of uniformly positive diameter. Annular estimates give a nonconstant locally uniform limit u and a finite measure μ such that

$$(\text{osc}_{\overline{B}(x,s)} u)^2 \leq C\mu(\overline{B}(x, C_\mu s)).$$

The approximants retain vanishing local squared oscillation sums after every *fixed* group transformation. A finite-cell inequality compares two such functions by sweeping through the separating levels. This is the analytic construction in Section 3.

The growth of the record moduli determines how we use this scalar limit. Exponential growth makes u locally Hölder, which allows normalized magnifications of its variation to converge to a nonconstant function on a punctured sphere. The level comparison forces each transform with a different puncture to be constant on the variation set of the first. A second magnification contradicts this constraint (Section 4).

For subexponential growth, we use the measure bound to distribute variation among many balls. A tree of separated balls carries nested value intervals and a probability flow. The transition probabilities obey linear and quadratic bounds in the ratio of child to parent interval lengths. These bounds force a definite total probability, at arbitrarily deep levels, onto nodes with two separated regions of variation (Section 5). At a common depth, many expansions have the same

two target regions but distinct punctures. Comparing them in pairs charges more mass to small balls than their overlap allows (Section 6). This excludes the remaining case.

Discrete crossing duality is part of the extremal-length theory developed in [22, 11]. The finite incidence-vector identity is also the probabilistic modulus formula of Albin–Poggi–Corradini [3, Theorem 5.1]; we reprove it in the notation of our cell covers. Pansu’s coarse conformal geometry uses image-diameter energies and condenser capacities [20, §6]. Scalar extremals and separating levels also enter Rajala’s uniformization of metric surfaces [21, Sections 4–6]. Here two path envelopes produce the required continua while preserving finite-cell oscillation bounds. The subsequent level integrations and pair counts are proved directly for the visual boundary. The two growth branches use this same scalar construction, and their combination finishes the theorem in Section 7.

Conventions. A path is a continuous map of a compact interval, identified with its image when convenient. A continuum is a nonempty compact connected set; it need not be path connected. $B(x, r)$ is open and $\overline{B}(x, r) = \{y : d(x, y) \leq r\}$ is closed. For real f , write $\text{osc}_A f = \sup_A f - \inf_A f$. Constants in $\lesssim$ and $\asymp$ are independent of the scale indices being varied. Dependence on a compact region, a fixed group element, or a subdivision parameter is specified where needed.

2 Boundary geometry and the modulus criterion

We collect the geometric estimates used throughout the proof, then state the classical uniformization input. Let Γ be the Cayley graph of G for a finite symmetric generating set, with unit edge lengths. Write e for its identity vertex and $|v| = d_\Gamma(e, v)$. The Gromov product is

$$(y|z)_o = \frac{1}{2}(d_\Gamma(o, y) + d_\Gamma(o, z) - d_\Gamma(y, z)).$$

Boundary products are defined by limits along rays, up to a uniformly bounded additive error. A visual metric d on $Z = \partial G$ satisfies

$$C_v^{-1}a^{-(y|z)_e} \leq d(y, z) \leq C_v a^{-(y|z)_e}, \quad y \neq z, \quad (2.1)$$

for some $a > 1$. Set $r_k = a^{-k}$ for integers $k \geq 0$. We use the standard product inequality $(y|z)_e \geq \min\{(y|w)_e, (w|z)_e\} - O(1)$ and its geometric form: geodesics with product at least $j - C$ stay a bounded synchronous distance apart through depth j , provided both have that length. The bound depends on C and the hyperbolicity constant. These facts, visual metrics, and thin ideal triangles are recalled in [9, III.H.3].

2.1 Expansions

Left multiplication by the inverse of a vertex far along a ray magnifies a small boundary ball. The following formula also controls the complement of that ball; this second feature will be essential.

Lemma 2.1 (Expansion formula). *Let v be the depth- k vertex on a based ray to $x \in Z$, put $h = v^{-1}$ and $R = r_k$, and set $D_y = \max\{R, d(x, y)\}$. There are constants $c_E, C_E > 0$, independent of these choices, such that*

$$c_E \frac{R d(y, z)}{D_y D_z} \leq d(hy, hz) \leq C_E \frac{R d(y, z)}{D_y D_z}. \quad (2.2)$$

In particular, on $B(x, CR)$ the map h is bi-Lipschitz with scale factor R^{-1} and distortion depending only on C . If $d(x, z_0) \geq \varepsilon_* > 0$, $R < \varepsilon_*/2$, and $p = hz_0$, then

$$d(hy, p) \lesssim R/D_y \quad (y \in Z), \quad d(hy, p) \asymp R/D_y \quad (d(x, y) \leq \varepsilon_*/2), \quad (2.3)$$

and

$$(v^{-1}|p)_e \geq k - C(\varepsilon_*). \quad (2.4)$$

The constants in the last two estimates may depend on ε_* .

Proof. Base change and thin triangles give

$$(y|z)_v = (y|z)_e + k - (v|y)_e - (v|z)_e + O(1), \quad (v|y)_e = \min\{k, (x|y)_e\} + O(1).$$

Thus $a^{-(v|y)_e} \asymp D_y$. Exponentiating the first identity and using $(hy|hz)_e = (y|z)_v + O(1)$ proves (2.2). With $z = z_0$, the factor D_{z_0} is bounded above and below, and $d(y, z_0)$ is bounded above. When $d(x, y) \leq \varepsilon_*/2$, the latter is also bounded below. This proves (2.3). Finally, $(v^{-1}|v^{-1}z_0)_e = k - (v|z_0)_e + O(1)$, and $(v|z_0)_e$ is bounded in terms of ε_* . $\square$

Lemma 2.2 (Prefixes). *Every vertex v lies a uniformly bounded distance from a depth- $|v|$ vertex on a based ray. With that ray's endpoint as x , (2.2) remains valid for $h = v^{-1}$ and $R = r_{|v|}$, with uniform constants. Moreover:*

- (i) *If $d(\xi, \eta) \leq Cr_j$, depth- j vertices on any based rays to ξ, η are a bounded distance apart, depending only on C .*
- (ii) *If $|v| = k$ and $(v|\xi)_e \geq k - C$, every geodesic from e to v stays a bounded synchronous distance from a ray to ξ through depth k .*

Proof. Translate a bi-infinite geodesic so that it passes through v . The ideal triangle with third vertex e puts v uniformly near one of its based sides; the nearby parameter differs from $|v|$ by a bounded amount. A bounded change of basepoint changes products by a bounded amount, preserving (2.2). The two remaining assertions follow from (2.1) and the product estimate. $\square$

Each fixed $g \in G$ acts bi-Lipschitzly on Z , with a constant depending on g . The whole action is uniformly quasi-Möbius: base-change factors cancel in distance cross-ratios, giving a single $C \geq 1$ such that

$$C^{-1} \frac{d(x, z)d(y, w)}{d(x, w)d(y, z)} \leq \frac{d(gx, gz)d(gy, gw)}{d(gx, gw)d(gy, gz)} \leq C \frac{d(x, z)d(y, w)}{d(x, w)d(y, z)}$$

for every distinct quadruple and every g .

2.2 Packing and paths

Lemma 2.3 (Packing). *There are $Q \geq 2$ and $C_P < \infty$ such that an s -separated subset of a ball of radius t has at most $C_P(t/s)^Q$ points whenever $0 < s \leq t \leq \text{diam } Z$. In particular, Z is doubling.*

Proof. Choose k with $r_k \asymp t$. Rays ending in the given ball have their depth- k vertices in a uniformly bounded graph ball, by Lemma 2.2. At depth $k + j$ there are at most CA^j possible continuations, where $A > 1$ depends on the valence. Endpoints whose rays share such a vertex are at visual distance at most $C'r_{k+j}$. Choose the least $j \geq 0$ with $C'r_{k+j} < s$. The desired bound follows with $Q \geq \max\{2, \log_a A\}$. Coarse scales are absorbed by increasing the constant. $\square$

Lemma 2.4 (Controlled paths). *There is $L \geq 1$ such that any two points of $\overline{B}(x, s)$ join by a path in $B(x, Ls)$. Any two points of a continuum can also be joined by a path in any prescribed open neighborhood of it.*

Proof. For small s , expand at x with $R \asymp s$ and choose z_0 a uniformly positive distance from x ; two fixed distinct points suffice for this choice. By (2.3), the two image points stay a uniform positive distance from $p = hz_0$. Fix a homeomorphism $\phi : Z \rightarrow \mathbb{S}^2$. Uniform continuity of ϕ^{-1} places their spherical images a uniform distance from $\phi(p)$. Join them outside a smaller round ball about $\phi(p)$. Uniform continuity of ϕ makes the pullback path stay a uniform d -distance from p . The upper bound in (2.3) puts its preimage under h in $B(x, Ls)$. Increase L for the remaining scales, using path connectedness of Z .

For a continuum C in an open set V , choose $\varepsilon > 0$ with its $L\varepsilon$ -neighborhood in V . Connectedness supplies a finite ε -chain in C between any two specified points: the set reachable by such chains is relatively clopen. Join consecutive points by the first assertion. $\square$

2.3 The classical reduction

Choose a maximal r_n -separated set $T_n \subset Z$ and its cells $B(q, 2r_n)$, $q \in T_n$. These cover Z ; fixed enlargements have bounded overlap. For a path family $\mathcal{F}$, define

$$\text{Mod}_2(\mathcal{F}, T_n) = \min_{\omega \geq 0} \left\{ \sum_{q \in T_n} \omega(q)^2 : \sum_{B(q, 2r_n) \cap \pi \neq \emptyset} \omega(q) \geq 1 \text{ for every } \pi \in \mathcal{F} \right\}.$$

Only finitely many incidence vectors occur, so the minimum exists. Fix a sufficiently small $d_0 > 0$ and put

$$\mathcal{F}_0 = \{\pi : \text{diam } \pi \geq d_0\}, \quad M(n) = \text{Mod}_2(\mathcal{F}_0, T_n). \quad (2.5)$$

We require $d_0 < \text{diam } Z/10$ and choose it so that every path of diameter at least $R/2$ in $\overline{B}(x, 3R)$ has image diameter at least d_0 under a scale- R expansion. This is possible by (2.2). We also impose the smallness requirement in Theorem 2.5 before fixing d_0 and $M(n)$. The family is nonempty, and $0 < M(n) < \infty$: all-one weights are admissible, while any one path gives the lower bound $1/\#T_n$.

Theorem 2.5 (Bonk–Kleiner; Bourdon–Kleiner). *If $\sup_n M(n) < \infty$, then (Z, d) is quasi-Möbius homeomorphic to the round sphere.*

Verification of the cited criterion. Corollary 3.5 of [8], derived from [6, Theorem 10.4], applies to an arcwise connected, approximately self-similar metric sphere. Approximate self-similarity means that each open ball, with distances divided by its radius, is uniformly bi-Lipschitz to an open subset of the space. Lemma 2.1 supplies this property; intermediate and large radii change only the constant.

The criterion assumes bounded 2-moduli of paths of diameter at least a sufficiently small d_0 in successive graph approximations. For a dyadic scale 2^{-m} choose n with $2^{-m}/a < r_n \leq 2^{-m}$. Our cells form a fixed κ -approximation for any $\kappa > 2a$: the inner balls of radius $2^{-m}/\kappa$ are disjoint, each is contained in its cell, and each cell is contained in the radius- $\kappa 2^{-m}$ ball about its center. The modulus is exactly $M(n)$. Finitely many coarse scales have finite modulus. No Coxeter-group assumption occurs in this criterion. $\square$

Proposition 2.6 (From the boundary to hyperbolic space). *If (Z, d) is quasi-Möbius homeomorphic to the round sphere, then G admits a proper cocompact isometric action on $\mathbb{H}^3$ with finite kernel.*

Proof. Let $\psi : Z \rightarrow \mathbb{S}^2$ be the uniformizing homeomorphism and write $\mathcal{G} = \{\psi g \psi^{-1} : g \in G\}$. The boundary cross-ratio estimate and the two fixed maps ψ, ψ^{-1} give one quasi-Möbius distortion function for every element of $\mathcal{G}$. Fix $f \in \mathcal{G}$ and $x \in \mathbb{S}^2$, keep a fourth point away from x , and let two points at equal spherical distance from x approach x . In the cross-ratio inequality the factors involving the fourth point tend to one, both before and after applying f . Thus the infinitesimal linear distortion has a bound independent of f and x ; the radius at which the estimate is accurate need not be uniform in f . The metric characterization of planar quasiconformality, applied in conformal charts, makes each f quasiconformal. At its almost-everywhere nonsingular differentiability points, the same infinitesimal bound controls the ratio of singular values of Df , and hence its analytic dilatation. This gives a common quasiconformal bound. For a reversing map apply the characterization after a conformal reflection; see [4, Theorems 2.5.3–2.5.4] and the metric characterization in [17, Theorem 1.4 and Remark 1.8(b)].

We recall explicitly the invariant-structure argument, including orientation reversal. This is the invariant-conformal-structure method of Sullivan, with the circumcenter construction detailed by Tukia [24, pp. 73–76]. Analytic quasiconformal maps have almost-everywhere defined nonsingular weak differentials, and they and their inverses preserve null sets [4, Theorem 3.1.2]. Their differentials satisfy the almost-everywhere composition rule [2, Lemma 10 and Theorem 5]. Since $\mathcal{G}$ is countable, remove the $\mathcal{G}$ -orbit of the union of all exceptional sets for these assertions and for all pairs of maps. Its complement X is invariant and conull. At $x \in X$, let $\mathcal{C}_x$ be the space of positive definite quadratic forms on $T_x \mathbb{S}^2$ modulo positive scalars. This is a hyperbolic plane, and pullback by any invertible linear map, of either orientation, is an isometry between the corresponding fibers [4, Section 2.3, equations (2.17)–(2.18) and Theorem 2.3.2]. The orbit $\{f^* q_{\text{round}}(x) : f \in \mathcal{G}\} \subset \mathcal{C}_x$ is uniformly bounded, by the common quasiconformal bound, and contains the round form. Its unique hyperbolic circumcenter defines a uniformly bounded measurable conformal structure $q(x)$. Indeed, circumcenters of the increasing finite subsets from an enumeration are measurable and converge to that circumcenter: their radii increase, their centers stay in a fixed compact ball, and every subsequential limit is the unique center of the whole orbit. The composition rule permutes this orbit under pullback, so $f^* q = q$ on X for every $f \in \mathcal{G}$. Extend q arbitrarily across the null complement of X .

The measurable Riemann mapping theorem [2, Theorem 6] gives a quasiconformal homeomorphism $F : \mathbb{S}^2 \rightarrow \mathbb{S}^2$ with $F^* q_{\text{round}} = q$ almost everywhere. Invariance and the analytic composition rule show that every $F f F^{-1}$ is conformal or anticonformal, according to the orientation of f . Such sphere homeomorphisms are Möbius or anti-Möbius. Their unique extensions to hyperbolic isometries therefore define a homomorphism $\rho : G \rightarrow \text{Isom}(\mathbb{H}^3)$.

For completeness, properness and cocompactness follow directly from the action on ordered distinct triples $\mathcal{T}(Z)$. On any compact set of triples all pairwise distances are bounded below. For $|v| \rightarrow \infty$, at least two points of such a triple stay away from the expansion center for v^{-1} ; their image distance is $O(r_{|v|})$. Thus $v^{-1} K \cap K = \emptyset$ eventually for every compact $K \subset \mathcal{T}(Z)$. Conversely, expand at one point of a closest pair with scale comparable to their distance. Formula (2.2) makes all three image distances uniformly positive: if the third point is nearby, all denominator factors have that scale; otherwise its distances to the pair cancel its denominator factor. Triples already uniformly separated need no expansion. Hence the action on triples is proper and cocompact.

These properties survive boundary conjugacy. Let $I = \text{Isom}(\mathbb{H}^3)$. The stabilizer H of a standard ordered boundary triple has two elements: the identity and reflection in its hyperbolic plane. Choose $o \in \mathbb{H}^3$ fixed by H , and let K_o be its compact stabilizer in I . The equivariant map

$$\mathcal{T}(\mathbb{S}^2) \cong I/H \longrightarrow I/K_o \cong \mathbb{H}^3$$

is continuous, onto, and proper: isometries sending o into a compact set form a compact set. Properness and cocompactness therefore descend. The kernel fixes o , so is finite. $\square$

3 A scalar limit from unbounded modulus

Assume that the macroscopic moduli $M(n)$ are unbounded. Crossing duality first gives transverse weights of energy $O(M(n)^{-1})$ along a subsequence. We turn those weights into continuous functions whose level continua meet every path straddling the corresponding value. An annular estimate then yields a nonconstant limit u and a finite measure controlling its squared oscillation. The approximating functions retain vanishing local oscillation energy after each fixed group transformation; later comparisons will use this property as well as the limit itself. Constants are independent of the approximation index; thresholds may depend on the fixed compact subset of the domain under consideration.

3.1 Record scales and crossing duality

If $\limsup M(n)^{1/n} > 1$, choose $1 < \lambda < \limsup M(n)^{1/n}$; otherwise put $\lambda = 1$. In both cases $M(n)/\lambda^n$ is unbounded. Retain increasing record indices N whose record values tend to infinity. Thus $M(N) \rightarrow \infty$ and

$$M(N - l) \leq \lambda^{-l} M(N) \quad (0 \leq l \leq N). \quad (3.1)$$

This inequality survives every further subsequence.

Lemma 3.1 (Localization to a rectangle). *There are coordinate rectangles $\mathcal{Q} = [0, 1]^2 \subset \widehat{\mathcal{Q}} = [0, 1] \times [-1, 2] \subset Z$ and a subsequence of the record indices such that the left-to-right crossing modulus of $\mathcal{Q}$ is at least $cM(N)$, with $c > 0$ independent of N .*

Proof. Cover Z by finitely many coordinate boxes $[-1, 1]^2$, each lying in a chart containing $[-8, 8]^2$ of diameter less than d_0 . A macroscopic path meeting an inner box must leave its larger box. Follow it to its first meeting with $\partial[-2, 2]^2$. If it exits through $x = 2$, its segment after the last preceding meeting with $x = 1$ crosses $[1, 2] \times [-2, 2]$ from left to right. The other three exit sides give analogous strips. Each strip extends by its height in both transverse directions inside $[-8, 8]^2$.

Consequently there are finitely many crossing families $\mathcal{F}_1, \dots, \mathcal{F}_J$ such that every macroscopic path contains a member of one of them. Summing admissible weights gives

$$M(N) \leq J \sum_{j=1}^J \text{Mod}_2(\mathcal{F}_j, T_N) \leq J^2 \max_j \text{Mod}_2(\mathcal{F}_j, T_N).$$

One strip realizes this bound along a subsequence. Affine coordinates give the asserted rectangles. $\square$

The edge $y = 0$ is the top of $\mathcal{Q}$. The rectangles $[0, 1] \times [-1, 0]$ and $[0, 1] \times [1, 2]$ are its top and bottom bands, respectively. Coordinates serve only interpolation and topology: all distances refer to d .

Lemma 3.2 (Crossings and blockers). *In a closed coordinate rectangle:*

- (i) *every continuum meeting both lateral edges meets every top-to-bottom path;*
- (ii) *every compact set meeting all top-to-bottom paths contains a continuum meeting both lateral edges.*

Proof. For (i), suppose a lateral continuum and a top-to-bottom path are disjoint. Enlarge the rectangle, extending the continuum horizontally and the path vertically to the corresponding new edges. The extensions remain disjoint: a new horizontal and vertical segment can meet only at an original corner, already a common point of the original sets. Approximate the extended path, within its positive clearance from the continuum, by a polygonal path with interior in the enlarged rectangle. After subdivision at intersections it contains a simple polygonal arc between the top and bottom edges. The Jordan curve theorem, applied after adjoining a boundary route, shows that this arc separates the lateral edges. The connected continuum cannot avoid it.

For (ii), subdivide into finer square grids and mark the closed squares meeting the blocker K . No edge-adjacent chain of unmarked squares joins top to bottom. Label A the unmarked squares reachable from the top through unmarked edge-adjacent squares, and append a top row of A squares. Label all remaining squares and an appended bottom row B . Draw the interfaces between unlike labels. Interior vertices have even degree; each lateral boundary has an odd number of ends, since its labels begin with A and end with B . There are no top or bottom ends. Some interface component therefore joins the lateral boundaries: otherwise every component contributes an even number of ends to the only lateral boundary it meets.

Along a path in this component, the adjacent B squares are original marked squares. An adjacent unmarked square would be reachable, while an interface with the appended bottom row would give an unmarked top-to-bottom chain. Successive B squares meet at least at a corner. Their union is a connected compact lateral crossing. A Hausdorff limit as the mesh tends to zero is connected, meets both lateral edges, and lies in K . $\square$

The next finite-dimensional argument belongs to discrete extremal length. Cannon–Floyd–Parry describe optimal weights through crossing vectors and their pairing with transverse cuts [11, Sections 2.3 and 2.4.3]; the probability formulation of Albin–Poggi–Corradini gives the same least-norm incidence-vector identity [3, Theorem 5.1]. We give the argument for the cell families used here.

Lemma 3.3 (Crossing dual weights). *For each retained N there are nonnegative weights $\rho_q = \rho_q^{(N)}$, $q \in T_N$, admissible for top-to-bottom crossings of $\mathcal{Q}$, such that*

$$\sum_{q \in T_N} \rho_q^2 \leq \frac{C}{M(N)}. \quad (3.2)$$

Proof. Let P minimize Euclidean norm in the convex hull of the finitely many cell-incidence vectors v of left-to-right crossings. Its coordinates are nonnegative and sum to at least one, so $P \neq 0$. Minimality gives $\langle P, v \rangle \geq \|P\|^2$, making $P/\|P\|^2$ admissible. Conversely any admissible ω satisfies $\langle \omega, P \rangle \geq 1$. Thus the left-to-right modulus is $\|P\|^{-2}$. By Lemma 3.2(i), each top-to-bottom incidence vector has inner product at least one with every v , hence with P . Set $\rho = P$ and apply Lemma 3.1. $\square$

3.2 Continuous functions with separating levels

For $\pi \subset \hat{\mathcal{Q}}$ write $\ell_\rho(\pi) = \sum_{q: B(q, 2r_N) \cap \pi \neq \emptyset} \rho_q$. For $x \in \mathcal{Q}$, let $b_N(x)$ be the infimum of this cost over paths from the top edge to x in $\mathcal{Q}$. Define the raw function $\min\{1, b_N\}$, assigning exactly zero on the top edge, and extend it by zero and one to the top and bottom bands. Admissibility gives value one on the bottom edge.

Along every path in $\hat{\mathcal{Q}}$, the difference of raw endpoint values is at most its cost. Inside $\mathcal{Q}$ this follows by concatenating with nearly minimizing paths and clipping at one; a path beginning on

the top edge is itself a competitor. A path visiting both bands contains a crossing of cost at least one. If it visits only the top band, each endpoint value is bounded by the cost of its segment to that band. If it visits only the bottom band, each endpoint value is at least one minus that segment's cost, by concatenation with a top-to-endpoint competitor. In either case the endpoint difference is bounded by the cost of the entire path.

Triangulate $\widehat{\mathcal{Q}}$ by a finite coordinate grid containing the band boundaries, with every triangle of d -diameter less than r_N . Interpolate the raw vertex values affinely to obtain a continuous $U_N^0 : \widehat{\mathcal{Q}} \rightarrow [0, 1]$. It is constant on the bands. Continuity alone does not ensure that one selected level continuum meets every straddling path. The following two path envelopes give the strict sublevel and superlevel sets the connections to the bands needed for that conclusion. Define

$$F_N(x) = \inf_{\pi: x \rightarrow \text{top}} \max_{\pi} U_N^0, \quad H_N(x) = \sup_{\pi: x \rightarrow \text{bottom}} \inf_{\pi} F_N,$$

where paths remain in $\widehat{\mathcal{Q}}$ and end in the indicated band.

Lemma 3.4 (Regularization and level continua). *The functions F_N, H_N are continuous, with the prescribed band values, and for every path $K \subset \widehat{\mathcal{Q}}$,*

$$\text{osc}_K H_N \leq \text{osc}_K F_N \leq \text{osc}_K U_N^0. \quad (3.3)$$

For every $0 < t < 1$, the set $\{H_N = t\} \cap \mathcal{Q}$ contains a lateral crossing continuum $C_{N,t}$ of diameter at least a fixed $d_1 > 0$. Every path in $\widehat{\mathcal{Q}}$ whose endpoint values strictly straddle t meets $C_{N,t}$.

Proof. We have $F_N \geq U_N^0$ and $H_N \leq F_N$; constant paths give the band values. Concatenation with nearly optimizing paths gives, for $x, y \in K$,

$$F_N(x) \leq \max\{\max_K U_N^0, F_N(y)\}, \quad H_N(y) \geq \min\{\inf_K F_N, H_N(x)\}.$$

If $F_N(x) > F_N(y)$, the first inequality places their values between $\min_K U_N^0$ and $\max_K U_N^0$. If $H_N(x) > H_N(y)$, the second places them between $\inf_K F_N$ and $\sup_K F_N$. This proves (3.3). Short coordinate segments and continuity of U_N^0 then prove continuity of both functions, including at the boundary.

Every point of $\{H_N > t\}$ joins the bottom band within that set: choose a path with $\inf F_N > t$ and use its tails. Likewise every point of $\{F_N < t\}$ joins the top within $\{F_N < t\}$. To connect a point x with $H_N(x) < t \leq F_N(x)$ to the top, choose $H_N(x) < s < t$ and follow any top-going path until its first point with $F_N = s$. On this prefix $F_N \geq s$, so concatenation with nearly maximizing paths from any prefix point z gives

$$H_N(x) \geq \min\{s, H_N(z)\}.$$

Hence $H_N(z) \leq H_N(x) < t$ throughout the prefix. Its endpoint joins the top through $F_N < t$, hence through $H_N < t$. Thus both strict sides of the level join their respective bands, and then the outer top and bottom edges, without meeting the level.

The compact set $\{H_N = t\}$ blocks all top-to-bottom paths by continuity and lies in $\mathcal{Q}$. By Lemma 3.2(ii) it contains a lateral continuum $C_{N,t}$; take d_1 to be the distance between the lateral edges of $\mathcal{Q}$. Complete any straddling path to a top-to-bottom path using the strict-side connections. The completed path meets $C_{N,t}$ by Lemma 3.2(i), and the added pieces avoid the level, proving the last assertion. $\square$

Figure 1 isolates the last step: the strict-side extensions cannot account for the intersection with the selected continuum, so the original straddling path must meet it.

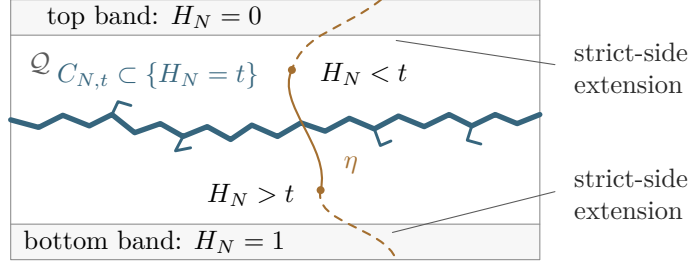


Figure 1: A separating level continuum and a straddling path. The dashed extensions stay in the strict sublevel and superlevel sets and complete η to a top-to-bottom crossing. That crossing meets $C_{N,t}$, so η itself meets $C_{N,t}$. The continuum is drawn schematically; neither smoothness nor path connectedness is assumed.

3.3 Oscillation and annular estimates

The level continua now let us estimate oscillation by crossing an annulus. Energies formed from image diameters and their relation to condenser capacity appear in Pansu's coarse conformal geometry [20, §6]; the estimates below use the finite-cell energy of our regularized scalar functions. Put $U = \text{int}_Z \tilde{Q}$. For a continuous function f on an open set containing a compact set K , and $J \geq 2$, define

$$\mathcal{E}_n^{(J)}(f; K) = \sum_{\substack{q \in T_n \\ B(q, 2r_n) \cap K \neq \emptyset}} (\text{osc}_{\overline{B}(q, Jr_n)} f)^2.$$

The closed balls lie in the domain for all sufficiently large n .

Lemma 3.5 (Local oscillation). *For fixed compact $K \subset U$ and $J > 0$, uniformly for $x \in K$ and sufficiently large N ,*

$$\text{osc}_{\overline{B}(x, Jr_N)} H_N \leq \sum_{\substack{q \in T_N \\ d(q, x) \leq C(J)r_N}} \rho_q. \quad (3.4)$$

Here $C(J)$ depends only on J and the ball path constant. Consequently, for every fixed $g \in G$, compact $K' \subset gU$, and $J \geq 2$,

$$\mathcal{E}_N^{(J)}(H_N \circ g^{-1}; K') \leq \frac{C(g, K', J)}{M(N)} \rightarrow 0.$$

Proof. Join two points of $\overline{B}(x, Jr_N)$ inside $B(x, L Jr_N)$ by Lemma 2.4. Vertices of triangles meeting this path lie in $B(x, (LJ + 1)r_N)$; any two such vertices join inside $B(x, L(LJ + 1)r_N) \subset U$ for large N . The raw endpoint inequality bounds every difference of their values by the right side of (3.4), with $C(J) = L(LJ + 1) + 3$. It therefore bounds the interpolant's oscillation along the original path. Apply (3.3).

Squaring and summing costs a bounded factor by Lemma 2.3: both the number of nearby weights and the number of centers charging each weight are bounded. The same holds for br_N -separated centers, with constants depending on $b > 0$. A fixed g is bi-Lipschitz, so inverse images of target balls lie in source balls of radius $C_g Jr_N$, their centers are $b_g r_N$ -separated, and all these source balls lie in a fixed compact subset of U . The summation bound and (3.2) finish the proof. $\square$

The mesh-scale estimate preserves the small dual energy. We next compare oscillation across a ball of radius $R = r_l$ with the dual energy in a surrounding annulus. Expansion takes that annulus to a fixed-size region and takes the mesh r_N to r_{N-l} ; this is why the next bound contains $M(N-l)$.

Lemma 3.6 (Annular estimate). *For every compact $K \subset U$ there are $R_* > 0$ and an integer m_0 such that, if $x \in K$, $R = r_l \leq R_*$, and $N-l \geq m_0$, then any two points joined by a path in $B(x, R/2)$ have H_N -value difference Δ satisfying*

$$\Delta^2 \leq CM(N-l) \sum_{\substack{q \in T_N \\ R/4 \leq d(q,x) \leq 4R}} \rho_q^2. \quad (3.5)$$

The constant is uniform; N is sufficiently large for the fixed interior region.

Proof. Choose R_* with $4R_* < d_1$ and all closed $5R_*$ -balls about K contained in U . All following restrictions on m_0 concern only $r_N/R = a^{-(N-l)}$.

Expand at x, l by h , and choose macroscopic admissible weights ω at scale $N-l$ with squared mass $M(N-l)$. Put

$$\mathcal{A} = \{q \in T_N : R/2 \leq d(q, x) \leq 3R\}, \quad \sigma_q = \sum_{\substack{\xi \in T_{N-l} \\ d(\xi, hq) \leq C_0 r_{N-l}}} \omega_\xi, \quad q \in \mathcal{A},$$

where $C_0 > C_E + 2$. Expansion on $B(x, 3R)$ separates the hq by a fixed multiple of r_{N-l} . Packing bounds both overlap multiplicities, giving

$$\sum_{q \in \mathcal{A}} \sigma_q^2 \leq CM(N-l).$$

Let I be the open interval between the endpoint values. For $t \in I$, the continuum $C_{N,t}$ meets the given path and exits $B(x, 2R)$, since its diameter exceeds $4R$. By Lemma 2.4, join these two continuum points in its $r_N/4$ -neighborhood and retain a radial subpath η_t between radii R and $2R$, contained in that closed annulus. Its diameter is at least R , so $h\eta_t$ is macroscopic by the choice of d_0 in Section 2.

If a target cell $B(\xi, 2r_{N-l})$ meets $h\eta_t$ at hy , choose $q \in T_N$ with $d(q, y) \leq r_N$. For large m_0 , this center belongs to $\mathcal{A}$ and $d(hq, hy) \leq C_E r_N/R$, so ω_ξ occurs in σ_q . Some level point lies within $r_N/4$ of y , hence in $\overline{B}(q, 2r_N)$. Thus t lies in the closed range interval I_q of H_N on that ball. Accounting for all target cells gives

$$1 \leq \sum_{q \in \mathcal{A}} \sigma_q \mathbf{1}_{I_q}(t) \quad (t \in I), \quad \Delta \leq CM(N-l)^{1/2} \left(\sum_{q \in \mathcal{A}} |I_q|^2 \right)^{1/2}.$$

The second inequality follows by integration and Cauchy–Schwarz. Apply (3.4) with $J = 2$ and its bounded-overlap summation. Taking $C(2)r_N < R/4$ confines every charged center to $R/4 \leq d(q, x) \leq 4R$, proving (3.5). $\square$

3.4 The limiting function and measure

The finite measures $\mu_N = M(N) \sum_{q \in T_N} \rho_q^2 \delta_q$ have uniformly bounded mass by (3.2).

Proposition 3.7 (Scalar limit). *Along a subsequence of the record indices, $H_N \rightarrow u$ locally uniformly on U and $\mu_N \rightarrow \mu$, where $u : U \rightarrow [0, 1]$ is continuous and nonconstant and μ is finite. The function u is zero and one on the interiors of the respective bands, and is locally Hölder continuous if $\lambda > 1$. For every compact $K \subset U$ and sufficiently small $s > 0$, uniformly for $x \in K$,*

$$(\text{osc}_{\overline{B}(x,s)} u)^2 \leq C\mu(\overline{B}(x, C_\mu s)). \quad (3.6)$$

For each $0 < t < 1$ there is a continuum $C_t \subset \mathcal{Q}$ of diameter at least d_1 , with $u = t$ on $C_t \cap U$, meeting every compact path in U whose endpoint values strictly straddle t . The fixed-transform energy bounds of Lemma 3.5 hold along the retained subsequence.

Proof. Fix compact $K \subset U$, and let $x, y \in K$ with $d(x, y) \leq r_k$. They join inside $B(x, Lr_k)$. Choose b with $a^b > 2L$ and consider indices $l \leq k - b$ above a fixed interior threshold, in an arithmetic progression of step b_1 with $a^{b_1} > 16$. The annuli $\{r_l/4 \leq d(x, \cdot) \leq 4r_l\}$ are disjoint. For N sufficiently large relative to k , (3.5) and (3.1) give

$$|H_N(x) - H_N(y)|^2 \leq C\lambda^{-l}\mu_N\{r_l/4 \leq d(x, \cdot) \leq 4r_l\}.$$

Summing over at least ck indices yields $|H_N(x) - H_N(y)| \leq C_K k^{-1/2}$. If $\lambda > 1$, the single choice $l = k - b$ gives $C_K \lambda^{-k/2}$ instead.

For each tolerance this controls a tail uniformly; the finitely many omitted functions are individually uniformly continuous. Thus the sequence is equicontinuous on K . Arzelà–Ascoli on a compact exhaustion of U , followed by weak compactness of the measures on Z , gives the asserted subsequence. The two constant band values prove nonconstancy. The stronger estimate gives local Hölder exponent $\log \lambda / (2 \log a)$.

For (3.6), fix $A > 2L$ and choose $As \leq R = r_l < aAs$. Any two points of $\overline{B}(x, s)$ join in $B(x, Ls) \subset B(x, R/2)$. The annular and record estimates give

$$(\text{osc}_{\overline{B}(x,s)} H_N)^2 \leq C\mu_N(\overline{B}(x, 4aAs))$$

for large N . Local uniform convergence passes the oscillation to the limit; the closed-set inequality for weak convergence bounds the right-hand limsup by $C\mu(\overline{B}(x, 4aAs))$. Take $C_\mu = 4aA$.

For each fixed t , take a Hausdorff limit C_t of $C_{N,t}$ in $\mathcal{Q}$. It is connected and has diameter at least d_1 . Local uniform convergence gives $u = t$ on $C_t \cap U$. A fixed compact straddling path also straddles t for every sufficiently large H_N , hence meets those $C_{N,t}$ and their limit. The subsequence used here may depend on t ; no simultaneous selection is needed. $\square$

3.5 Two comparison tools

The scalar construction will be used in three ways. The large continua C_t provide levels for annular constructions, with $u = t$ on $C_t \cap U$; no value of u is asserted outside U . The measure estimate (3.6) turns oscillation on a small ball into a lower bound for its enlarged ball's μ -mass. To compare transformations, we retain H_N and $C_{N,t}$: the identity $H_N = t$ holds on the entire continuum, and the local oscillation energies tend to zero after each transformation has been fixed. Those comparisons will pass to subcontinua contained in compact subsets of the common open domains.

The first tool integrates variation of a second function over a family of level continua. It is a finite-cell statement, independent of the preceding construction.

Lemma 3.8 (A finite-cell level comparison). *Let f, w be continuous on an open set $V \subset Z$, let $K \subset V$ be compact, and let $I \subset \mathbb{R}$ be an interval. Suppose that for every $t \in I$ there is a continuum $D_t \subset K$ with $f = t$ on D_t and $\text{osc}_{D_t} w \geq b_0 > 0$. For sufficiently large n ,*

$$b_0 |I| \leq \sum_{\substack{q \in T_n \\ B(q, 2r_n) \cap K \neq \emptyset}} \text{osc}_{\overline{B}(q, 2r_n)} f \text{osc}_{\overline{B}(q, 2r_n)} w. \quad (3.7)$$

Consequently these hypotheses cannot hold for sequences with both local oscillation energies on K tending to zero, if b_0 and $|I|$ stay bounded below by positive constants.

Proof. Let $I_q(f)$ be the closed range interval of f on the closed cell. For large n these cells lie in V . The interval $w(D_t)$ is covered by the range intervals of w on cells meeting D_t , and each such cell has $t \in I_q(f)$. Therefore

$$b_0 \leq \sum_{q: B(q, 2r_n) \cap K \neq \emptyset} \text{osc}_{\overline{B}(q, 2r_n)} w \mathbf{1}_{I_q(f)}(t).$$

Integrate over I and use $|I \cap I_q(f)| \leq \text{osc}_{\overline{B}(q, 2r_n)} f$. Only finitely many interval indicators are integrated, so no measurable selection of D_t is required. Cauchy–Schwarz gives the final assertion. $\square$

Lemma 3.9 (Stopping a continuum at a sphere). *Let D be a continuum, $p \in Z$, and $r > 0$. If $x \in D \setminus \overline{B}(p, r)$ and $D \cap B(p, r) \neq \emptyset$, then D contains a subcontinuum through x meeting the radius- r sphere and avoiding $B(p, r)$.*

Proof. In a compact metric space X , a point's component is the intersection Q_x of its clopen neighborhoods. To verify connectedness of Q_x , suppose it is separated into two compact pieces in disjoint open sets O_1, O_2 . Compactness of $X \setminus (O_1 \cup O_2)$ gives a finite intersection W of clopen neighborhoods of x contained in their union. Its intersection with the O_i containing x is clopen in X and misses the other piece of Q_x , a contradiction. The same compactness argument shows that every open neighborhood of a component contains a clopen neighborhood of it.

Apply this to $X = D \setminus B(p, r)$ and the component C of x . If C missed the sphere, it would have a clopen neighborhood W in X missing the sphere. Compactness places W a positive distance from $\overline{B}(p, r)$, so W is also clopen in D . It is nonempty and misses $D \cap B(p, r)$, contradicting connectedness. Thus C is the required subcontinuum. $\square$

4 Exponential records: comparing two magnifications

Suppose $\lambda > 1$. The scalar limit u is then locally Hölder. We magnify its variation to a nonconstant function on a punctured sphere. The approximating level continua force every transform with a different puncture to be constant wherever this function varies. A second magnification makes two such transforms vary on the same two disks, with disjoint value ranges, yielding a contradiction. Scalar approximation indices throughout belong to the subsequence of [Proposition 3.7](#).

4.1 Magnifying variation

Lemma 4.1 (Hölder magnification). *Let f be a real-valued, locally Hölder continuous function on an open set $\Omega \subset Z$. Suppose O is open, $\overline{O} \subset \Omega$, and O contains a compact path on which f*

is nonconstant. Fix z_0 with $\text{dist}(z_0, O) > 0$. There are depths $k_i \rightarrow \infty$, centers $x_i \in O$, positive numbers σ_i , and expansions $h_i = v_i^{-1}$ at (x_i, k_i) such that, after passage to a subsequence,

$$p_i = h_i z_0 \longrightarrow b, \quad \frac{\text{dist}(x_i, Z \setminus O)}{r_{k_i}} \longrightarrow \infty.$$

Every compact subset of $Z \setminus \{b\}$ eventually lies in $h_i \Omega$, and

$$F_i(z) = \frac{f(h_i^{-1}z) - f(x_i)}{\sigma_i} \tag{4.1}$$

converges locally uniformly there to a nonconstant locally Hölder continuous function F .

Proof. Write $\delta(x) = \text{dist}(x, Z \setminus O)$ and fix $\eta > Q$. A finite cover of $\overline{O}$ gives a uniform Hölder bound, with some exponent $\alpha_0 > 0$, on a neighborhood of $\overline{O}$. For all large k put

$$A_k = \sup_{x \in O} \delta(x)^\eta \text{osc}_{\overline{B}(x, r_k)} f.$$

The balls lie in Ω , and

$$ca^{-Qk} \leq A_k \leq Ca^{-\alpha_0 k}. \tag{4.2}$$

For the lower bound, cover a compact variation path $P \subset O$ by at most Cr_k^{-Q} radius- r_k balls centered on P , using [Lemma 2.3](#). Their range intervals cover the interval $f(P)$, so one ball has oscillation at least cr_k^Q ; also $\min_P \delta > 0$. The upper bound is the Hölder estimate.

Fix $0 < \alpha < \alpha_0$. Since $A_k a^{\alpha k} \rightarrow 0$, there are arbitrarily late indices attaining their maximum over the remaining tail. At these indices,

$$A_{k+j} \leq a^{-\alpha j} A_k \quad (j \geq 0). \tag{4.3}$$

Choose $x_k \in O$ attaining at least half the supremum, and set $\delta_k = \delta(x_k)$ and $\sigma_k = A_k / \delta_k^\eta$. Boundedness of f and (4.2) give $\delta_k^\eta \geq cr_k^Q$, hence $\delta_k / r_k \geq cr_k^{Q/\eta-1} \rightarrow \infty$. Expand at (x_k, k) and take a subsequence with $p_k = h_k z_0 \rightarrow b$.

For a compact $K \subset Z \setminus \{b\}$, (2.3) gives

$$h_k^{-1}K \subset \overline{B}(x_k, C_K r_k) \subset O \tag{4.4}$$

eventually. The expansion formula gives $d(h_k^{-1}z, h_k^{-1}w) \leq C_K r_k d(z, w)$ on K . On every fixed multiple of this source ball, δ is comparable to δ_k . Thus (4.3) implies

$$\frac{|f(y) - f(y')|}{\sigma_k} \leq Ca^{-\alpha j} \quad \text{if } d(y, y') \leq r_{k+j}, \quad j \geq 0. \tag{4.5}$$

For sufficiently small $d(z, w)$, choosing $j \geq 0$ with $a^{-j} \geq C_K d(z, w) > a^{-j-1}$ proves a uniform local α -Hölder estimate for F_k on K .

For boundedness, [Lemma 2.4](#) joins $h_k^{-1}z$ to x_k inside a fixed multiple of $B(x_k, r_k)$. Packing covers this path by a bounded number of radius- r_k balls centered on the path, each with oscillation at most $C\sigma_k$. Their range intervals cover the path's image, so $|F_k(z)| \leq C_K$. Compact exhaustion and Arzelà–Ascoli now give a locally uniform, locally Hölder limit. Finally, $\text{osc}_{h_k \overline{B}(x_k, r_k)} F_k \geq 1/2$. The lower puncture bound places all these image balls in one compact set off b , so local uniform convergence makes the limit nonconstant. $\square$

4.2 A common mesh and escaping level continua

Apply [Lemma 4.1](#) to u on U , taking a small variation region O and z_0 outside its closure. Call the limit W , its puncture b , and retain x_i, k_i, h_i, σ_i .

Lemma 4.2 (Diagonal approximants). *After thinning the expansions, there are increasing scalar indices N_j such that*

$$f_j(z) = \frac{H_{N_j}(h_j^{-1}z) - u(x_j)}{\sigma_j}, \quad z \in h_j \widehat{\mathcal{Q}},$$

converges locally uniformly to W on $Z \setminus \{b\}$. For every fixed $g \in G$, compact $K \subset Z \setminus \{gb\}$, and fixed $J \geq 2$,

$$\mathcal{E}_{N_j}^{(J)}(f_j \circ g^{-1}; K) \longrightarrow 0. \quad (4.6)$$

All indicated enlarged balls eventually lie in the functions' domains.

If a compact path $P \subset Z \setminus \{b\}$ has endpoint values of W strictly straddling a compact positive-length interval I , then, for all large j and every $t \in I$, there is a continuum $D_{j,t} \subset h_j \widehat{\mathcal{Q}}$ with

$$f_j = t \text{ on } D_{j,t}, \quad D_{j,t} \cap P \neq \emptyset, \quad \sup_{t \in I} \text{dist}(b, D_{j,t}) \longrightarrow 0.$$

Proof. For $j \geq 4$ put $K_j = \{d(z, b) \geq 1/j\}$ and $L_j = \{d(z, b) \geq 1/(2j)\}$. Choose the j th expansion so that $h_j^{-1}L_j \subset U$ and its normalized u -transform differs from W by at most $1/j$ on L_j . Hold this expansion and its normalization fixed. Then choose $N_j > N_{j-1}$ so large that

$$\sup_{L_j} \frac{|H_{N_j}(h_j^{-1}z) - u(h_j^{-1}z)|}{\sigma_j} \leq \frac{1}{j}, \quad \mathcal{E}_{N_j}^{(j)}(f_j; K_j) \leq \frac{1}{j}, \quad (j+2)r_{N_j} < \frac{1}{2j}.$$

The uniform approximation follows from [Proposition 3.7](#). With h_j and σ_j fixed, [Lemma 3.5](#) bounds the displayed energy by $C(h_j, K_j, j)/(\sigma_j^2 M(N_j))$, so the energy condition follows by increasing N_j . The last condition keeps all the enlarged balls in L_j . This proves local uniform convergence and vanishing energy at every fixed enlargement off b .

For a fixed g , choose a compact neighborhood K' of $g^{-1}K$ avoiding b . Pulling back each target ball and choosing a nearest source net point gives, by bi-Lipschitz distortion and packing,

$$\mathcal{E}_n^{(J)}(F \circ g^{-1}; K) \leq C_g \mathcal{E}_n^{(J')}(F; K')$$

for some fixed $J' \geq 2$ and all large n . Each source center receives only boundedly many target centers. Eventually $K' \subset K_j$ and $J' \leq j$, proving (4.6).

Uniform convergence makes the endpoint values of f_j straddle I . The source level $\tau_j(t) = u(x_j) + \sigma_j t$ therefore lies in $(0, 1)$, and [Lemma 3.4](#) supplies $D_{j,t} = h_j C_{N_j, \tau_j(t)}$, meeting P . Its source continuum has diameter at least d_1 , so contains a point at distance at least $d_1/2$ from x_j . By (2.3),

$$\text{dist}(b, D_{j,t}) \leq d(b, p_j) + Cr_{k_j} \longrightarrow 0$$

uniformly for $t \in I$. □

4.3 Comparison across punctures

Let $\mathcal{N}(F)$ be the points of a continuous function's domain at which it is not locally constant.

Lemma 4.3 (Transform comparison). *If $c = gb \neq b$ and $V = W \circ g^{-1}$, then*

$$V(x) = V(b) \quad (x \in \mathcal{N}(W) \setminus \{c\}). \quad (4.7)$$

More generally, for $W_s = W \circ s^{-1}$ and $p_s = sb$, if $p_s \neq p_t$,

$$W_t(x) = W_t(p_s) \quad (x \in \mathcal{N}(W_s) \setminus \{p_t\}). \quad (4.8)$$

Proof. Suppose $x \in \mathcal{N}(W) \setminus \{c\}$ and $V(x) \neq V(b)$. Choose $\varepsilon > 0$ and a coordinate disk about x , compactly contained off b, c , on which $|V - V(b)| \geq 4\varepsilon$. It contains a compact variation path P for W . Choose a compact positive-length interval I strictly between its endpoint values, avoiding $W(c)$. Continuity gives $\epsilon_c > 0$ such that $\overline{B}(c, \epsilon_c)$ avoids b and its W -range has positive distance from I . Consequently, for all large j , the continua $D_{j,t}$ from Lemma 4.2 avoid this closed ball.

Choose $\rho > 0$ so that $\overline{B}(b, \rho)$ avoids c, P and $|V - V(b)| \leq \varepsilon$ there. For $w_j = f_j \circ g^{-1}$, local uniform convergence gives

$$|w_j - V(b)| \geq 3\varepsilon \quad \text{on } P, \quad |w_j - V(b)| \leq 2\varepsilon \quad \text{on } \overline{B}(b, \rho)$$

for all large j . Each $D_{j,t}$ meets P and enters $B(b, \rho)$. By Lemma 3.9, it has a subcontinuum joining a chosen point of P to the radius- ρ sphere and contained in

$$K = \{z : d(z, b) \geq \rho, d(z, c) \geq \epsilon_c\}.$$

This subcontinuum lies in the level $f_j = t$ and has w_j -oscillation at least ε . A fixed compact neighborhood of K avoiding both punctures eventually lies in $h_j U \cap gh_j U$ and contains every closed radius- $2r_{N_j}$ cell meeting K . Hence Lemma 3.8 and Cauchy–Schwarz give

$$\varepsilon|I| \leq (\mathcal{E}_{N_j}^{(2)}(f_j; K))^{1/2} (\mathcal{E}_{N_j}^{(2)}(w_j; K))^{1/2} \longrightarrow 0,$$

a contradiction. All sets and g are fixed in this limit. Finally apply (4.7) at $s^{-1}x$ with group element $s^{-1}t$ to obtain (4.8). $\square$

4.4 The second magnification

We first control repeated punctures. An accumulation direction of vertices going to infinity is a subsequential boundary limit.

Lemma 4.4 (Stabilizer directions). *For every $b \in Z$, the vertices of $E = \text{Stab}_G(b)$ have at most two accumulation directions.*

Proof. Let ξ be a based ray to b . For $q \in E$, $|q| \leq t$, thinness of the ideal triangle with sides $[e, q], \xi, q\xi$ puts $q\xi(n)$ uniformly near ξ when n is sufficiently larger than t . The nearby parameter lies in $[n - t - C, n + t + C]$. For one common integer n , these vertices are distinct as q varies. Bounded valence therefore gives

$$\#\{q \in E : |q| \leq t\} \leq C'(t + 1). \quad (4.9)$$

If another accumulation direction exists, take $q_n \rightarrow x \neq b$ and $q_n^{-1} \rightarrow y$ along a subsequence. Base change gives, uniformly on compact sets avoiding y ,

$$(q_n | q_n z)_e = |q_n| - (q_n^{-1} | z)_e + O(1) \geq |q_n| - O(1).$$

By Lemma 2.2, $q_n z \rightarrow x$ there, and $q_n b = b$ forces $y = b$. Choose disjoint small neighborhoods of b, x , with a point outside their union. For large n , $q = q_n$ sends the complement of the first into the second. Its positive iterates never return the outside point, so q has infinite order. It is therefore loxodromic, with endpoints b, x' ; see [9, III.Γ.3.10].

If some $h \in E$ moved x' , the cyclic subgroups generated by q and $h q h^{-1}$ would intersect trivially: a nonzero common power would have the same pair of endpoints. Their m^2 products $q^i (h q h^{-1})^j$, $1 \leq i, j \leq m$, would then be distinct and have length $O(m)$, contradicting (4.9). Thus every element of E fixes both endpoints. It maps a geodesic line between them to a line at uniformly bounded Hausdorff distance, so its orbit of e stays a fixed distance from that line. Its only possible accumulation directions are the two endpoints. $\square$

Proposition 4.5. *The contradiction setup cannot have $\lambda > 1$.*

Proof. Let $E = \text{Stab}(b)$ and let Λ_E be its at-most-two accumulation directions. Choose a variation region O for W whose closure avoids $\{b\} \cup \Lambda_E$. Such a region exists: otherwise W would be locally constant on the connected sphere minus this finite set, and continuity would make it constant throughout its domain.

Apply Lemma 4.1 to W on $Z \setminus \{b\}$, with this O and $z_0 = b$. Write the expansions as $s_i = v_i^{-1}$, with centers $y_i \in O$, depths $\ell_i \rightarrow \infty$, and punctures $p_i = s_i b \rightarrow b_*$. Positive affine normalizations of $W_i = W \circ s_i^{-1}$ converge locally uniformly off b_* to a continuous nonconstant W_* .

The punctures have a pairwise distinct subsequence. Indeed, if $v_i^{-1} b = v^{-1} b$ infinitely often, then $v_i v^{-1} \in E$. After a subsequence $y_i \rightarrow y \in \overline{O}$. Since v_i lies at depth ℓ_i on a ray to y_i , Lemma 2.2 gives $v_i \rightarrow y$ in boundary direction. Bounded right multiplication preserves this direction, implying $y \in \Lambda_E$, a contradiction.

Choose two coordinate disks D_1, D_2 , with compact closures off b_* , having disjoint W_* -range intervals and each containing a compact variation path. To do this, follow a variation path to its first hitting points of two distinct intermediate values. Just before either first hitting time the values differ, so the function is not locally constant at either point. Continuity supplies disks with disjoint ranges, and non-local constancy supplies a variation path in each.

For every sufficiently large i , the puncture p_i is outside the closed disks. Uniform convergence of the normalized W_i preserves both the disjoint range intervals and the positive variation on the two fixed paths. Positive affine normalization preserves these properties, so $\mathcal{N}(W_i)$ meets each disk. Fix two distinct such indices i, j . By (4.8), W_j has the single value $W_j(p_i)$ at every point of $\mathcal{N}(W_i)$ in either disk. Taking one point in each contradicts the disjoint W_j -range intervals. $\square$

5 A probability flow in the subexponential case

Assume now that the unbounded moduli satisfy

$$\limsup_{j \rightarrow \infty} M(j)^{1/j} \leq 1. \quad (5.1)$$

Let u, μ, C_t be given by Proposition 3.7. We build a tree of small balls carrying nested intervals of values of u . At each node, either two separated variations survive expansion, or many children carry a substantial fraction of its interval. A probability flow will force the first alternative at arbitrarily deep levels.

Choose $x_0 \in U$ where u is not locally constant, and choose $\epsilon_0 > 0$ with $\overline{B}(x_0, 3\epsilon_0) \subset U$. Put

$$K = \overline{B}(x_0, \epsilon_0), \quad K^+ = \overline{B}(x_0, 2\epsilon_0),$$

and, shrinking ϵ_0 if necessary, choose z_0 a positive distance from K^+ . All centers will lie in K and all source neighborhoods in K^+ . Use the constants of (3.6) for this fixed region.

5.1 Node intervals and transition probabilities

The tree will record both where variation occurs and which values it carries. A node has a source center x , a scale R , and a value interval J . Its children have scale $r = \theta R$, where $\theta = a^{-D}$ for one positive integer subdivision depth D used throughout the tree. Their intervals are contained in J . We will choose probabilities on the children that control interval length linearly and quadratically; the two controls have different roles in the final comparison.

We first define this data for geometric constants $c_0 > 2L + 2$, $Lc_0 + 2 < A < B_0 - 2$, and $H > 0$. Here c_0 is the source-ball enlargement, A, B_0 locate an annulus of possible child centers, and Hr is the separation required of retained children. The next subsection fixes these constants, before choosing D . We use only scales $R = r_k$ small enough that all fixed enlargements below lie in K^+ and satisfy (3.6), and that $(B_0 + 2)R < d_1/2$.

A *node* is a center $x \in K$, a scale $R = r_k$, and a nonempty open interval $J = (j_-, j_+) \subset (0, 1)$ such that

$$J \subset \left(\min_{\overline{B}(x, c_0 R)} u, \max_{\overline{B}(x, c_0 R)} u \right). \quad (5.2)$$

Write $\Delta = |J|$. For every $t \in J$, a path in $B(x, Lc_0 R)$ strictly straddles t and hence meets C_t . Since $\text{diam } C_t \geq d_1$, the continuum leaves $\overline{B}(x, (B_0 + 2)R)$ and meets every intervening radial sphere.

Take a maximal r -separated set Y in

$$\mathcal{A}_x(R) = \{y : (A - 1)R \leq d(x, y) \leq (B_0 + 1)R\}.$$

It is an r -net there, and packing gives

$$\#Y \leq C_Y a^{QD}, \quad (5.3)$$

with C_Y independent of D . For $y \in Y$ let I_y^{in} and I_y^{out} be the closed intervals between the extrema of u on $\overline{B}(y, 2r)$ and $\overline{B}(y, c_0 r)$, respectively. These are range *intervals*; the balls need not be connected. Put

$$J_y = J \cap \text{int } I_y^{\text{out}}, \quad t_y = |J_y|/\Delta.$$

A positive t_y gives a nested open child interval satisfying (5.2), and, including singleton ranges,

$$|J \cap I_y^{\text{out}}| = t_y \Delta. \quad (5.4)$$

For $0 < c_* \leq 1/(8 \max\{C_Y, 1\})$, put $\kappa = c_* a^{-QD}$. A node is *good* if two candidates y^-, y^+ satisfy

$$|I_{y^-}^{\text{in}} \cap (j_-, j_- + \Delta/4)| \geq \kappa \Delta, \quad |I_{y^+}^{\text{in}} \cap (j_- + 3\Delta/4, j_+)| \geq \kappa \Delta, \quad (5.5)$$

$$\text{dist}(I_{y^-}^{\text{out}}, I_{y^+}^{\text{out}}) \geq \Delta/4. \quad (5.6)$$

Goodness uses the full candidate set, independently of the children later retained.

The required probabilities are as follows. Fix the exponents

$$b_* = \frac{1}{2}, \quad \tau = \frac{1}{20}, \quad \beta = \frac{\tau}{4(Q+1)}, \quad \gamma = \frac{\beta}{2}. \quad (5.7)$$

At every node we seek an Hr -separated set of children, with probabilities summing to one, such that

$$p_y \leq a^{-\gamma D} t_y, \quad p_y \leq a^{-b_* D} t_y^2 \quad \text{at nodes that are not good,} \quad (5.8)$$

$$p_y \leq a^{-\gamma D} t_y, \quad p_y \leq a^{\tau D} t_y^2 \quad \text{at good nodes.} \quad (5.9)$$

Multiplying the linear bounds along a branch will telescope the interval ratios and make the branch probability small. Multiplying the quadratic bounds will compare probability to squared interval length. The measure estimate bounds the sum of those squares at each separated generation. It will therefore bound the average probability of nodes with the stronger quadratic bound and force a substantial amount of good-node probability at arbitrarily deep levels.

Two facts will supply these transitions. A node that is not good has many children with $t_y \geq 1/8$, on which uniform probabilities suffice. At every node, a capacity estimate supplies enough total weight to satisfy the weaker pair of bounds (5.9). After constructing the transitions, we show that goodness also gives two separated regions of variation in one of finitely many target patterns. That geometric property is the input to [Section 6](#).

5.2 Geometric choices and many large children

We now fix the geometry used by the definitions above. Put $c_{\text{in}} = 2L$ and choose $c_0 > c_{\text{in}} + 2$ so large that, with

$$A = 4Lc_0 + 20, \quad B_0 = A + 10, \quad c_{\text{pat}} = \frac{2C_E c_{\text{in}}}{(A - 2)^2},$$

we have

$$\frac{c_E c_0}{(B_0 + 2)^2} > 12c_{\text{pat}}. \quad (5.10)$$

The constant c_{pat} will be the target radius factor: paths joining extrema of the two inner child balls will expand into balls of radius $c_{\text{pat}}\theta/2$, while the larger source balls will contain the preimages of radius- $10c_{\text{pat}}\theta$ target balls. The ratio of the left side of (5.10) to the right side grows linearly in c_0 , so this choice is possible. Next choose

$$H > \max\{4(B_0 + 1), 4C_\mu c_0, 2\}.$$

The first inequality preserves separation between different branches; the second makes the measure enlargements at one generation disjoint. These choices fix C_Y in (5.3), and we fix c_* as specified above. Require $\theta < 1/4$ and $c_0\theta < 1$. All further lower bounds on D will depend only on this fixed data.

The uniform upper bound on node scales can be chosen now, before D : all source radius factors are fixed, and the subsidiary scales r and s used below never exceed R . Thus the source neighborhoods and measure enlargements stay in K^+ for every node. We choose the root scale after D has been fixed.

Lemma 5.1 (Many children at a node that is not good). *For sufficiently large D , every node that is not good has at least $c_H a^D$ mutually Hr -separated candidates with $t_y \geq 1/8$, where $c_H > 0$ is independent of the node and D .*

Proof. Fix $b \in [AR, B_0R]$. Each C_t , $t \in J$, meets the radius- b sphere, so the inner intervals of candidates with $|d(x, y) - b| \leq r$ cover J . By (5.3), two of them meet the respective quarters in (5.5) by at least $\kappa\Delta$.

If both have $t_y < 1/8$, the lower-quarter outer interval ends below $j_- + 3\Delta/8$, and the upper-quarter interval starts above $j_- + 5\Delta/8$. Indeed each meets its quarter but overlaps J in length less than $\Delta/8$; the assertion also holds for intervals extending beyond J . Their distance exceeds $\Delta/4$, making the node good. Otherwise one candidate has $t_y \geq 1/8$. Apply this at radial values spaced by $(H+3)r$: there are at least $c_H a^D$ such values, and the resulting centers are more than Hr apart. $\square$

At a node that is not good, put uniform probabilities on the candidates in [Lemma 5.1](#). For large D they satisfy (5.8): indeed, $p_y \leq c_H^{-1} a^{-D}$ and $t_y \geq 1/8$, while $\gamma, b_* < 1$.

5.3 The capacity estimate

It remains to construct transitions satisfying (5.9). A single subdivision depth D must work even when a node's interval J is arbitrarily short. We therefore integrate over J before choosing D ; its length will cancel from the estimate.

Lemma 5.2 (Separated capacity). *For sufficiently large D , uniformly over all nodes, there is an Hr -separated subset $Y' \subset Y$ such that*

$$\sum_{y \in Y'} \min\{a^{-\gamma D} t_y, a^{\tau D} t_y^2\} \geq 1. \quad (5.11)$$

Proof. Join two candidates in a graph when their distance is at most Hr . Packing bounds its degree, so graph coloring partitions Y into at most C_H separated classes, with C_H independent of D . If the conclusion fails, then

$$\sum_{y \in Y} \min\{a^{-\gamma D} t_y, a^{\tau D} t_y^2\} < C_H. \quad (5.12)$$

Call y *large* when $t_y \geq a^{-(\tau+\gamma)D}$ and *small* otherwise. On large candidates the minimum is linear, so

$$\sum_{\text{large } y} t_y \leq C_H a^{\gamma D}. \quad (5.13)$$

We will test levels at an intermediate radius s , larger than the child radius r and smaller than the parent radius R . This provides more radial tests than the large ranges can occupy. At the same time, it lets us detect the remaining levels with a modulus cost at the relative scale r/s .

Remove the large ranges. Set $D' = \lceil \beta D \rceil$ and $s = a^{-D'} R$. For large D , $0 < D' < D$ and $s < R/4$. Choose s -spaced radial bins $\mathcal{B}$ in $[(A+1)R, (B_0-1)R]$, so $\#\mathcal{B} \asymp a^{D'}$. On $\mathcal{B} \times J$ use counting measure times Lebesgue measure. Mark (b, t) if a large candidate satisfies

$$|d(x, y) - b| \leq 5s, \quad t \in I_y^{\text{out}}.$$

Each candidate meets at most 11 bins. By (5.4) and (5.13), the marked measure is at most $CC_H a^{\gamma D} \Delta$. Since $\gamma < \beta$, the unmarked measure is at least

$$ca^{D'} \Delta. \quad (5.14)$$

Detect the remaining levels by modulus. Take a maximal s -separated set $\mathcal{Z}_s$ in $\mathcal{A}_x(R)$; its radius- s balls cover that annulus, and it has at most $Ca^{QD'}$ points. Expand at each $z \in \mathcal{Z}_s$

at depth $k + D'$ by h_z . Put $q = r/s = a^{-(D-D')}$, and take macroscopic admissible weights ω_ξ , $\xi \in T_{D-D'}$, of squared mass $M(D - D')$. For fixed $C_T > 2 + C_E$, define

$$\sigma_y^z = \begin{cases} \sum_{\xi: d(\xi, h_z y) \leq C_T q} \omega_\xi, & d(y, z) \leq 3s, \\ 0, & d(y, z) > 3s, \end{cases} \quad \sigma_y = \sum_{z \in \mathcal{Z}_s} \sigma_y^z.$$

Within $\overline{B}(z, 3s)$ the candidate images are separated by a fixed multiple of q . Thus every sum has boundedly many terms and every target weight is charged boundedly many times. Thus $\sum_y (\sigma_y^z)^2 \leq CM(D - D')$ for each z . Also each y lies within $3s$ of boundedly many auxiliary centers. Packing and Cauchy–Schwarz give

$$\sum_y \sigma_y^2 \leq Ca^{QD'} M(D - D'). \quad (5.15)$$

Fix an unmarked (b, t) . Choose $w \in C_t$ on the radius- b sphere and $z \in \mathcal{Z}_s$ with $d(w, z) \leq s$. The diameter bound gives a point of C_t outside $\overline{B}(w, 2s)$. Join these points in the $r/4$ -neighborhood of C_t , by [Lemma 2.4](#), and stop at the first radius- s hit about w . The resulting path π has diameter at least s and lies in $\overline{B}(w, s) \subset \overline{B}(z, 2s)$. The scale- s expansion h_z therefore makes it macroscopic by the choice of d_0 in [Section 2](#).

The radial bin lies between $(A + 1)R$ and $(B_0 - 1)R$, while $s < R/4$. Thus every point of π lies between radii $(A + 3/4)R$ and $(B_0 - 3/4)R$ about x . In particular it lies in the candidate annulus and has a candidate center within distance r . The same fixed source margin places every level point within $r/4$ of π in $K^+ \subset U$. On these points the scalar-limit property gives $u = t$, even though it does not assert this identity on parts of C_t outside U .

For a target cell meeting $h_z \pi$, choose a preimage hit $v \in \pi$ and $y \in Y$ with $d(y, v) \leq r$. Then $d(y, z) < 3s$. If ξ is the target cell's center, [\(2.2\)](#) gives $d(\xi, h_z y) < (2 + C_E)q < C_T q$, so its weight occurs in σ_y^z . A point of C_t is within $r/4$ of v , hence within $5r/4 < c_0 r$ of y . The preceding interior check gives $t \in I_y^{\text{out}}$. Since $|d(x, y) - b| \leq s + r < 5s$, unmarkedness makes y small. Admissibility therefore yields

$$\sum_{\text{small } y: |d(x, y) - b| \leq 5s} \sigma_y \mathbf{1}_{I_y^{\text{out}}}(t) \geq 1 \quad \text{on every unmarked pair.} \quad (5.16)$$

The unmarked levels now have an admissible detection weight supported on small candidates. We compare its squared mass with the surviving amount of level parameter.

Compare the energies. Integrating [\(5.16\)](#), using bounded bin multiplicity, gives

$$ca^{D'} \Delta \leq C \Delta \sum_{\text{small } y} \sigma_y t_y.$$

Only finitely many closed-interval indicators are integrated; no measurable selection of continua or paths is needed, and singleton ranges contribute zero. Cancel $\Delta > 0$. Cauchy–Schwarz and [\(5.15\)](#) imply

$$\sum_{\text{small } y} t_y^2 \geq \frac{ca^{-(Q-2)D'}}{M(D - D')}. \quad (5.17)$$

Subexponential growth gives $M(D - D') \leq a^{\tau D/4}$ for large D : first bound $M(j) \leq Ca^{\tau j/8}$ and then absorb C by increasing D . Since $D' \leq \beta D + 1$ and $(Q - 2)\beta < \tau/4$,

$$\sum_{\text{small } y} a^{\tau D} t_y^2 \geq ca^{[3\tau/4 - (Q-2)\beta]D} \geq ca^{\tau D/2}.$$

On small candidates this is exactly the capacity, contradicting (5.12). Every threshold for D is uniform over nodes. $\square$

Fix such a D . At each good node, divide the positive capacities on a set Y' from Lemma 5.2 by their sum. These probabilities satisfy (5.9). At the remaining nodes use the uniform rule above. Every positive transition retains a nonempty open child interval $J_y \subset J$.

5.4 Common target patterns

The probabilities are now fixed. We verify the additional geometric information carried by a good node: its two source variations expand into a pair of balls chosen from a finite list, with separated range intervals. The list depends on the fixed D , but not on the node's generation or interval length.

Fix a node's expansion $h = v^{-1}$ when constructing it. At a good node, join the extrema of each inner ball by a path in $B(y^\pm, c_{\text{in}}r)$. Each path has endpoint variation at least $\kappa\Delta$. Its points lie at distance at least $(A-2)R$ from x , so (2.2) puts its image within $c_{\text{pat}}\theta/2$ of $hy^\pm$. Conversely, write $D_w = \max\{R, d(x, w)\}$. If $d(y, z) \geq c_0r$, then $D_z \leq D_y + d(y, z)$ and monotonicity of $s/(D_y + s)$ give

$$d(hy, hz) \geq \frac{c_ERc_0r}{D_y(D_y + c_0r)} \geq \frac{c_EC_0\theta}{(B_0 + 2)^2} > 12c_{\text{pat}}\theta.$$

Thus

$$\overline{B}(hy, 10c_{\text{pat}}\theta) \subset h(B(y, c_0r)). \quad (5.18)$$

Choose a fixed finite $c_{\text{pat}}\theta$ -net in Z . For each $hy^\pm$ choose a net point within $c_{\text{pat}}\theta$ and take the closed radius- $3c_{\text{pat}}\theta$ ball about it. The ordered pair $(\mathcal{B}^-, \mathcal{B}^+)$ is the node's *pattern*. Each ball contains its path image and lies in the corresponding outer source-ball image. Consequently, for $u \circ h^{-1}$ the two whole-ball range intervals have distance at least $\Delta/4$, and each ball contains a path with endpoint variation at least $\kappa\Delta$. There are finitely many patterns, independently of depth.

5.5 The tree and its flow

Choose k_0 so large that $R_0 = r_{k_0}$ satisfies all the source-scale conditions above and

$$2(B_0 + 1)R_0 < \epsilon_0/2, \quad (B_0 + 2)R_0 < d_1/2.$$

Non-local-constancy at x_0 supplies a nonempty open interval $J_0 \subset (0, 1)$ strictly inside the range on $\overline{B}(x_0, c_0R_0)$. Write $\Delta_0 = |J_0|$ and construct the tree recursively, storing each node's expansion and, at good nodes, its pattern.

A child moves by at most $(B_0 + 1)R$ from a parent of scale R . Consequently all descendants stay within

$$\frac{B_0 + 1}{1 - \theta}R \leq 2(B_0 + 1)R \quad (5.19)$$

of that parent, and all centers lie in K . Generation n has scale $r_{k_0 + Dn}$ and its centers are H times that scale apart: this holds for siblings, while children of distinct parents of scale R are separated by at least

$$[H - 2(B_0 + 1)]R > HR/2 > H\theta R.$$

At any level of scale R , (3.6) bounds Δ_i^2 by $C\mu(\overline{B}(x_i, C_\mu c_0R))$. These enlarged balls are disjoint because $H > 4C_\mu c_0$, and hence

$$\sum_i \Delta_i^2 \leq E_0, \quad E_0 = C\mu(Z) < \infty. \quad (5.20)$$

Give the root probability 1 and let P_i be the product of transitions to node i . Every level has total probability 1. If i has generation n , let m_i count its ancestors at generations $0, \dots, n-1$ that are not good. Multiplying the transition bounds and telescoping the interval ratios yields

$$P_i \leq a^{-\gamma D n} \frac{\Delta_i}{\Delta_0}, \quad \Delta_i^2 \geq \Delta_0^2 P_i a^{D((b_* + \tau)m_i - \tau n)}. \quad (5.21)$$

Conditionally, if i is m generations below j , nesting gives

$$\frac{P_i}{P_j} \leq a^{-\gamma D m} \frac{\Delta_i}{\Delta_j} \leq a^{-\gamma D m}. \quad (5.22)$$

Lemma 5.3 (Levels with substantial good mass). *Infinitely many generations have total probability at least $1/2$ on good nodes.*

Proof. By (5.21), (5.20), and Jensen's inequality,

$$\frac{E_0}{\Delta_0^2} \geq \sum_i P_i a^{D((b_* + \tau)m_i - \tau n)} \geq a^{D((b_* + \tau) \sum_i P_i m_i - \tau n)}.$$

Thus

$$\frac{1}{n} \sum_i P_i m_i \leq \frac{\tau}{b_* + \tau} + O(1/n) = \frac{1}{11} + O(1/n).$$

Conservation of probability identifies the left side with the average bad mass over the first n generations. If good mass were eventually less than $1/2$, that average would have lower limit at least $1/2$, a contradiction. $\square$

Proposition 5.4 (Properties of the flow). *The construction gives a fixed D and a tree with generation- n centers $Hr_{k_0 + Dn}$ -separated and descendant displacement (5.19). Its nonempty value intervals are nested, its probabilities satisfy (5.21) and (5.22), and infinitely many levels have good mass at least $1/2$. Every node has a fixed expansion. Each good node has a pattern from a finite list of ordered target-ball pairs. For that expansion, each target ball contains a path with endpoint variation at least $\kappa \Delta_i$, their whole-ball range intervals have distance at least $\Delta_i/4$, and the balls lie in the images of the respective outer balls of radius $c_0 \theta r_{k_0 + Dn}$.*

6 Same-depth expansions and the measure comparison

We compare expansions at many nodes of a single generation. Almost every pair of punctures is sufficiently separated. A level-continuum argument then forces one of the two charts to spend a definite amount of oscillation near the other's puncture. Pulling those neighborhoods back gives more total measure than their overlap permits.

Proposition 6.1. *The moduli $M(n)$ cannot be unbounded while satisfying*

$$\limsup_{n \rightarrow \infty} M(n)^{1/n} \leq 1.$$

We prove this proposition through the pair comparison and counting estimates below. Fix the flow of Proposition 5.4, including D , and a compact neighborhood of its centers in U . Use the point z_0 chosen in the flow construction, a positive distance from that neighborhood. At generation n write

$$k = k_0 + Dn, \quad R = a^{-k}, \quad p_i = h_i z_0,$$

where $h_i = v_i^{-1}$ is the expansion already chosen for node i . The centers x_i are HR -separated, and

$$P_i \leq a^{-\gamma Dn} \frac{\Delta_i}{\Delta_0}, \quad \frac{P_j}{P_A} \leq a^{-\gamma Dm} \quad \text{if } j \text{ lies } m \text{ generations below } A. \quad (6.1)$$

We use the arbitrarily large generations having good mass at least $1/2$, supplied by [Lemma 5.3](#).

A good node has a pattern $(\mathcal{B}^-, \mathcal{B}^+)$ from a fixed finite list. Each target ball contains a compact path with endpoint $u \circ h_i^{-1}$ -variation at least $\kappa \Delta_i$, and lies in the image of its outer source ball. The two target range intervals have distance at least $\Delta_i/4$. Throughout, a *range interval* means the interval between the minimum and maximum; metric balls need not be connected.

All geometric and counting constants below are independent of the generation, nodes, and approximating index N . Constants and thresholds in convergence statements may depend on a pair after it has been fixed.

6.1 A probability class and separated punctures

Lemma 6.2 (Bounded puncture multiplicity). *For every fixed $C > 0$, the balls $\{\overline{B}(p_i, CR) : i \text{ at generation } n\}$ have uniformly bounded multiplicity.*

Proof. Centers assigned to the same depth- k vertex have mutual distance $O(R)$. Their HR -separation and [Lemma 2.3](#) bound their number. If several puncture balls share a point, their punctures have mutual products at least $k - O_C(1)$. By (2.4) and [Lemma 2.2](#), all corresponding inverse vertices lie in one bounded Cayley-graph ball. There are boundedly many such vertices, each with the bounded multiplicity just proved. $\square$

There are at most Ca^{Qk} nodes. Discarding those with $P_i < a^{-2Qk}$ loses probability at most Ca^{-Qk} , so the remaining good nodes have mass at least $1/3$ for large n . Partitioning by pattern and into the $O(k)$ dyadic probability classes gives a class $\mathcal{I}$ with

$$P_* \leq P_i \leq 2P_* \quad (i \in \mathcal{I}), \quad \sum_{i \in \mathcal{I}} P_i \geq \frac{c}{k}. \quad (6.2)$$

Writing $\mathcal{N} = |\mathcal{I}|$, we have

$$\mathcal{N}P_* \geq \frac{c}{2k}. \quad (6.3)$$

Also $P_i \leq a^{-\gamma Dn}$ by (6.1), so $\mathcal{N} \geq ck^{-1}a^{\gamma Dn} \rightarrow \infty$. We will use punctures separated at a scale a^{-l} much larger than R . The final source-overlap estimate carries a factor a^{Ql} , so we arrange $Ql \leq \gamma Dn$. Put

$$\zeta = \gamma/Q, \quad m = \lfloor \zeta n \rfloor, \quad l = Dm.$$

Both l and $k - l$ tend to infinity.

Lemma 6.3 (Probability near a puncture). *For every generation- n node i ,*

$$\sum_{\substack{j \text{ at generation } n \\ d(p_j, p_i) \leq a^{-l}}} P_j \leq Ca^{-\gamma Dm}. \quad (6.4)$$

Proof. Fix an ancestor A at generation $n - m$, whose depth is $k - l = k_0 + D(n - m)$. We claim that only boundedly many descendants j have $d(p_j, p_i) \leq a^{-l}$. Split a geodesic word

$$v_j^{-1} = w_j u'_j, \quad |w_j| = l, \quad |u'_j| = k - l.$$

The puncture condition and (2.4) give $(p_j|p_i)_e \geq l - C$ and $(v_j^{-1}|p_j)_e \geq k - C$. Thus w_j lies a bounded distance from the depth- l point on a fixed ray to p_i , leaving boundedly many possibilities.

Reversing and inverting *the same word* gives the geodesic $v_j = (u'_j)^{-1}w_j^{-1}$. Its depth- $(k - l)$ prefix is therefore $(u'_j)^{-1}$. Since

$$d(x_j, x_A) \leq 2(B_0 + 1)r_{k-l},$$

thinness and Lemma 2.2 put this prefix a bounded distance from the depth- $(k - l)$ point on a fixed ray to x_A . Hence u'_j also has boundedly many possibilities. Their products determine boundedly many vertices v_j , and the first step of Lemma 6.2 bounds the nodes assigned to each. We count group elements here, so nonuniqueness of geodesic spellings causes no additional multiplicity. The source and inverse prefixes together account for the full word.

Each such descendant has $P_j \leq P_A a^{-\gamma D_m}$ by (6.1). Sum first over these descendants and then over ancestors, whose probabilities sum to one. $\square$

For each $i \in \mathcal{I}$, the fraction of its class with puncture distance at most a^{-l} is bounded by

$$\frac{C a^{-\gamma D_m}}{\mathcal{N} P_*} \leq C k a^{-\gamma D_m} = o(1).$$

This includes the diagonal. Consequently at least $c\mathcal{N}^2$ unordered pairs satisfy

$$d_{ij} := d(p_i, p_j) > a^{-l}. \quad (6.5)$$

Call them *far pairs*. Uniformly for these pairs, $R/d_{ij} \leq a^{-(k-l)} \rightarrow 0$.

6.2 Comparison on a compact common domain

Let $K \subset U$ contain all centers, with its closed ϵ -neighborhood still in U . The upper puncture bound gives

$$d(h_i y, p_i) \geq A_h R/2 \implies d(x_i, y) \leq D_y \leq 2C/A_h. \quad (6.6)$$

Choose A_h so that $2C/A_h < \epsilon/2$ and $A_h/2 > 2C/d_1$, and set

$$E_i = \overline{B}(p_i, A_h R).$$

Every $h_i C_{N,t}$ reaches $B(p_i, A_h R/2)$: the diameter bound supplies a source point at distance at least $d_1/2$ from x_i , whose image has puncture distance at most $2CR/d_1$.

The target balls lie a uniform distance $\delta_* > 0$ from p_i . Indeed, their preimages lie in the outer source balls, where $D_y \leq (B_0 + 2)R$, so the lower puncture bound applies for large n . Because the pattern is shared, this holds for every $i \in \mathcal{I}$. For large n , uniformly for far pairs,

$$2A_h R < a^{-l}, \quad A_h R < \delta_*/2.$$

Thus E_i, E_j are disjoint, each lies compactly in the other's transformed domain by (6.6), and the pattern balls avoid both. These thresholds are independent of N .

Lemma 6.4 (Pair alternative). *Every far pair at a sufficiently large generation has an endpoint i satisfying*

$$\text{osc}_{h_i^{-1}E_j} u \geq \frac{\kappa}{16} \Delta_i. \quad (6.7)$$

Proof. Write $F_i = u \circ h_i^{-1}$ and $F_j = u \circ h_j^{-1}$ on their open domains. If $\text{osc}_{E_i} F_j \geq \Delta_j/16$, endpoint j qualifies. Here we use $\kappa \leq 1$. Otherwise, one pattern ball $\mathcal{B}^s$ has its F_j range interval at distance at least $\Delta_j/16$ from that on E_i . Indeed, if neither did, the two pattern intervals would have distance less than $3\Delta_j/16$, contradicting their separation by $\Delta_j/4$.

Node i supplies a compact path $\eta \subset \mathcal{B}^s$ with endpoint F_i -gap at least $\kappa\Delta_i$. Suppose $\text{osc}_{E_j} F_i < \kappa\Delta_i/4$. Removing this latter range interval from the endpoint gap leaves a positive-length closed interval $I \subset (0, 1)$ strictly between the endpoint values and a positive distance from the range on E_j . To obtain positive margins, choose a smaller closed interval inside one of the remaining open intervals. Its length need not be uniform over pairs.

Fix the pair and I before letting $N \rightarrow \infty$. Put $F_{i,N} = H_N \circ h_i^{-1}$ and $F_{j,N} = H_N \circ h_j^{-1}$. The fixed range margins and local uniform convergence imply, for every $t \in I$ and all sufficiently large N , that $h_i C_{N,t}$ hits η and avoids E_j . It also reaches the inner i -puncture ball. For the first assertion, the endpoint values still strictly straddle all of I , so Lemma 3.4 applies to the preimage path. For the second, the range on E_j remains disjoint from I . By Lemma 3.9, it has a subcontinuum $K_{N,t}$ containing a hit on η , meeting the sphere $d(z, p_i) = A_h R$, and avoiding $B(p_i, A_h R)$. It remains in the level $F_{i,N} = t$ and avoids E_j . In particular,

$$K_{N,t} \subset K_{ij} := Z \setminus (B(p_i, A_h R) \cup B(p_j, A_h R)) \Subset h_i U \cap h_j U.$$

Compact containment follows from (6.6) with half the radius. The sphere hit belongs to E_i ; no identification of that sphere with the boundary of a metric ball is needed.

Uniform convergence on $\eta \cup E_i$ preserves the F_j range gap, giving $\text{osc}_{K_{N,t}} F_{j,N} \geq \Delta_j/32$ uniformly for $t \in I$. The finite-cell comparison, Lemma 3.8, now gives

$$\frac{\Delta_j}{32} |I| \leq \sum_{\substack{q \in T_N \\ B(q, 2r_N) \cap K_{ij} \neq \emptyset}} \text{osc}_{\overline{B(q, 2r_N)}} F_{i,N} \text{osc}_{\overline{B(q, 2r_N)}} F_{j,N}.$$

For large N these closed cells lie in the common domain. Both squared oscillation sums tend to zero by Lemma 3.5, so Cauchy–Schwarz contradicts the positive left side. Therefore $\text{osc}_{E_j} F_i \geq \kappa\Delta_i/4$, proving the alternative. The argument applies separately to every pair above the geometric threshold; no convergence rate uniform over pairs is required. $\square$

6.3 Cost balls and total multiplicity

We now turn the pair alternative into a measure bound and control the overlap of the sets being charged. The two indices have distinct roles: with i fixed, the target punctures control the number of partners j ; at a source point, separation of the centers controls the possible indices i . Both estimates will apply to the enlarged balls required by (3.6).

For an ordered far pair put $z_{ij} = h_i^{-1} p_j$. The relevant source scales are

$$\left. \begin{array}{l} \text{puncture denominator } D_{z_{ij}} \\ \text{radius of the charged neighborhood} \end{array} \right| \asymp \begin{array}{l} R/d_{ij} \\ R^2/d_{ij}^2. \end{array}$$

The radius divided by the denominator is $O(R/d_{ij}) \rightarrow 0$. This separation of scales is what allows a fixed enlargement of the charged neighborhood without losing control of its image.

If $y \in h_i^{-1} E_j$, then $d(h_i y, p_i) \geq d_{ij} - A_h R \geq d_{ij}/2$, whence $D_y \leq CR/d_{ij}$. These points approach x_i uniformly, so the two-sided puncture bound applies and gives

$$D_{z_{ij}} \asymp R/d_{ij}. \tag{6.8}$$

The lower expansion bound and $d(h_i y, p_j) \leq A_h R$ then give $d(y, z_{ij}) \leq CR^2/d_{ij}^2$. Fixing a sufficiently large C_2 , put $s_{ij} = C_2 R^2/d_{ij}^2$; thus

$$h_i^{-1} E_j \subseteq \overline{B}(z_{ij}, s_{ij}). \quad (6.9)$$

Choose the enlargement constant C_μ in (3.6) uniformly on a compact neighborhood of the centers and define

$$\mathcal{D}_{ij} = \overline{B}(z_{ij}, C_\mu s_{ij}).$$

Their centers stay in this neighborhood and their radii tend uniformly to zero. If i satisfies Lemma 6.4, (3.6) gives

$$\mu(\mathcal{D}_{ij}) \geq c\kappa^2 \Delta_i^2. \quad (6.10)$$

The enlarged balls still have small images. Indeed, D_y is 1-Lipschitz and

$$\frac{C_\mu s_{ij}}{D_{z_{ij}}} \leq C \frac{R}{d_{ij}} \leq C a^{-(k-l)} \rightarrow 0.$$

Thus $D_y \asymp D_{z_{ij}}$ on $\mathcal{D}_{ij}$, and the upper expansion bound yields

$$h_i \mathcal{D}_{ij} \subseteq \overline{B}(p_j, C_3 R). \quad (6.11)$$

Figure 2 shows the charged neighborhood and the image control that survives the fixed enlargement.

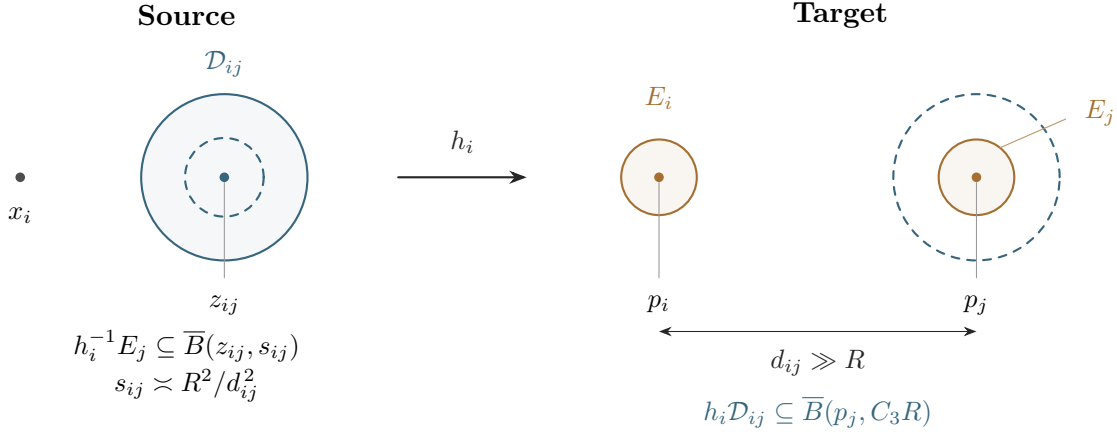


Figure 2: The measure cost of a far pair. The dashed source ball contains $h_i^{-1} E_j$ and has radius $s_{ij} \asymp R^2/d_{ij}^2$. Its fixed enlargement $\mathcal{D}_{ij}$, drawn solid, still maps into the dashed target ball of radius $C_3 R$. Here $p_i = h_i z_0$, $z_{ij} = h_i^{-1} p_j$, and E_i, E_j are the solid puncture balls of radius $A_h R$. The circles are schematic; no connectedness or boundary regularity of metric balls is assumed.

For fixed i , Lemma 6.2 therefore bounds the multiplicity of $\mathcal{D}_{ij}$ as j varies. Also

$$\mathcal{D}_{ij} \subseteq \overline{B}(x_i, C_4 R a^l), \quad (6.12)$$

because $d(x_i, z_{ij}) \leq CR/d_{ij}$ and the cost radius is $o(R/d_{ij})$. The HR -separation of the x_i and Lemma 2.3 bound the multiplicity of these larger balls by $C a^{Ql}$.

Conclusion of the proof of Proposition 6.1. Assign each unordered far pair to a qualifying endpoint. At any point, (6.12) allows only Ca^{Ql} paying indices i , and (6.11) allows boundedly many partners for each. Integrating this multiplicity bound gives

$$\sum_{\{i,j\} \text{ far, assigned to } i} \mu(\mathcal{D}_{ij}) \leq Ca^{Ql} \mu(Z). \quad (6.13)$$

This counts the enlarged balls directly and uses no invariance of μ . The assignment may be arbitrary when both endpoints qualify: all lower bounds hold for either endpoint, and the multiplicity estimate already counts every possible ordered contribution.

Conversely, every paying endpoint satisfies

$$\Delta_i^2 \geq \Delta_0^2 a^{2\gamma Dn} P_i^2 \geq \Delta_0^2 a^{2\gamma Dn} P_*^2.$$

There are at least $c\mathcal{N}^2$ far pairs, so (6.10) and (6.3) give total cost

$$c\mathcal{N}^2 \kappa^2 \Delta_0^2 a^{2\gamma Dn} P_*^2 \geq c' k^{-2} a^{2\gamma Dn}. \quad (6.14)$$

Since $\kappa, \Delta_0 > 0$ are fixed and $Ql = QD \lfloor \gamma n / Q \rfloor \leq \gamma Dn$, comparison with (6.13) yields

$$a^{\gamma Dn} \leq Ck^2.$$

This contradicts $k = k_0 + Dn$, excluding the unbounded subexponential case. $\square$

7 Completion of the proof

Proof of Theorem 1.1. If $M(n)$ were unbounded, the record selection of Section 3 would give either exponential records, excluded by Proposition 4.5, or subexponential growth, excluded by Proposition 6.1. Thus $\sup_n M(n) < \infty$. Theorem 2.5 uniformizes the boundary, and Proposition 2.6 supplies the proper cocompact action with finite kernel. $\square$

8 Consequences

Corollary 8.1. *If G is a torsion-free hyperbolic group and ∂G is homeomorphic to $\mathbb{S}^2$, then G is the fundamental group of a closed hyperbolic three-manifold.*

Proof. Point stabilizers in a proper action are finite, hence trivial when G is torsion-free. Thus the action in Theorem 1.1 is free, and $\mathbb{H}^3/G$ is a closed hyperbolic three-manifold with fundamental group G . The manifold may be nonorientable. $\square$

Corollary 8.2 (Virtual structure in the torsion-free case). *Let G be a torsion-free hyperbolic group with ∂G homeomorphic to $\mathbb{S}^2$. Some finite-index subgroup of G is isomorphic to $\pi_1(\Sigma) \rtimes \mathbb{Z}$ for a closed connected orientable surface Σ . In particular, G has positive virtual first Betti number. Moreover, G is virtually compact special and residually finite.*

Proof. Let M^+ be the orientation cover of the closed hyperbolic manifold in Corollary 8.1, using the manifold itself when it is orientable, and let $G^+ = \pi_1(M^+)$. Thus G^+ is normal of index at most two in G . Agol's virtual-fiber theorem [1, Theorem 9.2] gives a connected finite cover N of M^+ fibering over the circle with closed connected orientable fiber Σ . The fibration sequence

gives $\pi_1(N) \cong \pi_1(\Sigma) \rtimes \mathbb{Z}$, with action induced by the monodromy, and its surjection onto $\mathbb{Z}$ gives $b_1(\pi_1(N)) \geq 1$.

Bergeron–Wise’s cubulation theorem [5, Theorem 5.3] gives a proper cocompact action of G^+ on a CAT(0) cube complex, free because G^+ is torsion-free. Agol’s virtual-special theorem [1, Theorem 1.1] then gives a finite-index subgroup of G^+ with compact special quotient, so G is virtually compact special. Agol’s separability corollary [1, Corollary 1.3], applied to the trivial subgroup, also makes G^+ residually finite. The finite quotient G/G^+ and intersections of the finitely many G -conjugates of finite-index normal subgroups of G^+ then separate every nontrivial element of G . $\square$

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